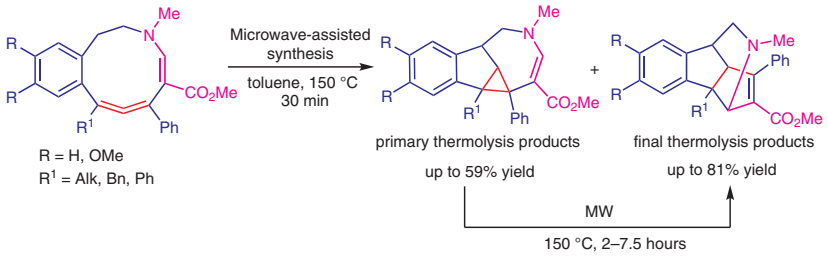
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Unusual Transformations of Cyclic Allenes with an Enamine Moiety into Complex Frameworks

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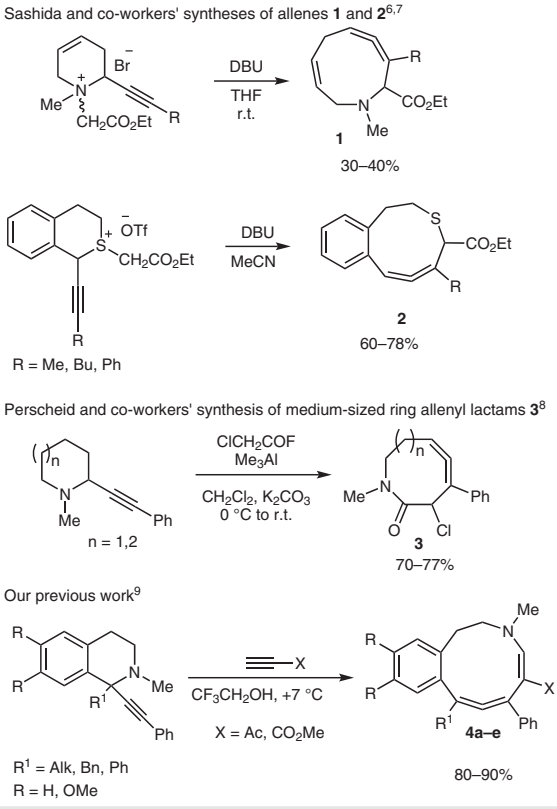
Abstract The thermolysis reactions of benzoazacyclodecatrienes under microwave irradiation conditions in toluene at 150 °C afforded complex azabenzo[*a*]cyclopropa[*cd*]azulene and (epiminomethano)cyclopenta[*a*]indene frameworks. Cyclopropanes were established as intermediates of the ultimate thermolysis products.

Key words thermal transformation, azacyclic allene, cyclopropanes, epiminomethanocyclopentaindenes, microwave-assisted reaction, polyheterocyclic compounds

In recent years, microwave irradiation has been widely applied in the synthesis of heterocyclic compounds, including polycyclic nitrogen-containing derivatives.[1–3] These approaches permit the efficient formation of structurally complex compounds that are interesting for their natural origin or their biological activity.[3–5] In this work, to increase the accessibility of complex frameworks, we proposed to use the thermal rearrangement of cyclic allenes containing an enamine fragment.

Synthetic methods for transformations of cyclic allenes, particularly those containing a heteroatom, remain largely unstudied. In selected examples, [2,3]- and [3,3]-sigmatropic rearrangements are crucial for the formation of heterocyclic allenes and their thermal transformation products (Scheme 1). In this manner, azacyclononatrienes **1** were obtained in low yields (30–40%) from 6-ethynyl-1-carboxymethyl-1,2,5,6-tetrahydropyridinium salts in the presence of DBU[6] (Scheme 1).

The same authors applied this method to a moderate-yielding construction of the sulfur-containing nonacyclic



Scheme 1 Examples of syntheses of heterocyclic allenes

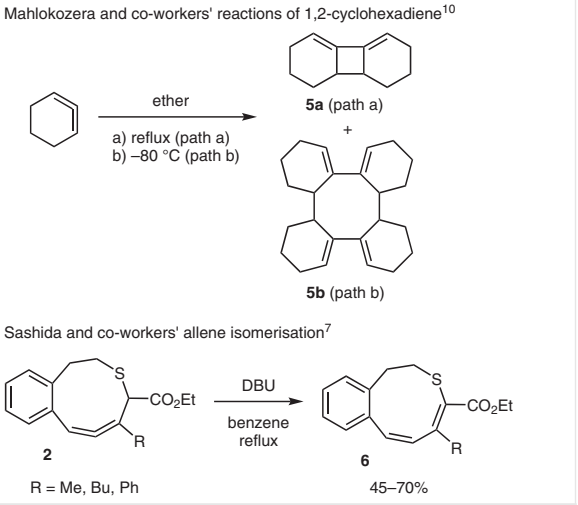
allenes **2** from the corresponding 1-alkynyl-2-(2-ethoxycarbonyl)-3,4-dihydro-1*H*-isothiochromenium triflates in the presence of DBU.[7] Lactams such as the 3-chloro-1-aza-deca- and -undeca-4,5-diene-2-ones **3** were obtained

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through a [3,3]-sigmatropic rearrangement of 2-(phenylethynyl)-*N*-methylpiperidine or -azepine, respectively, with chloroacetyl fluoride.[8] Our research group has also reported a synthesis of 8-substituted benzazecines **4** containing an allene moiety from 1-substituted 1-(phenylethynyl)-1,2,3,4-tetrahydroisoquinolines and electron-deficient alkynes (Scheme 1).[9]

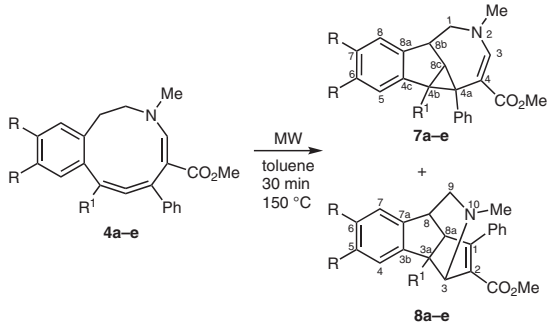
These first steps taken in obtaining cyclic allenes containing a ring heteroatom have permitted us to begin studying the properties of these compounds. Dimerization or intermolecular cycloaddition result in carbocycles in a reaction that is a characteristic chemical property of cyclic allenes.[10–12] The direction of this interaction is often determined by the temperature at which the reaction occurs (Scheme 2, upper). However, there is only one example of a thermal isomerization of a heterocyclic allene: the transformation of 1,2,4,-trihydro-3-benzothionines **2** into the corresponding 1,3-diene structures **6** (Scheme 2, lower).[7]



Scheme 2 Examples of thermal transformations of cyclic allenes

Our goal was to study the thermal transformation of azacyclic allenes **4a–e** (Table 1) and the effects of substituents on the thermolysis of these compounds. Azacyclodecatrienes **4a**, **4c**, and **4e** have been previously obtained by the reaction of the corresponding 1-(phenylethynyl)isoquinolines with methyl propiolate in trifluoroethanol at 7 °C (Scheme 1):[9] compounds **4b** and **4d** are novel.[13] Initially, benzoazacyclodecatrienes **4a–e** were heated in toluene under microwave irradiation conditions [150 °C, 30 min; see Supporting Information (SI) and Procedure A][14] to give mixtures of 2-azabenzo[*a*]cyclopropa[*cd*]azulenes **7a–e** and (epiminomethano)cyclopenta[*a*]indenes **8a–e**. Prolonged heating of azacyclic allenes **4a–e** in toluene under microwave irradiation caused the reaction to proceed to completion, yielding the (epiminomethano)cyclopenta[*a*]indenes **8a–e** exclusively. During this thermolysis process, monitoring by thin-layer chromatography showed that the starting allene **4** and cyclopropane **7** disappeared as indene **8** appeared. We therefore suggest that the cyclopropanes **7a–e** were the primary thermolysis products, and that further heating in toluene led to the formation of indenes **8a–e** (Table 1; see also SI and Procedure B).[15]

Table 1 Thermolysis of Allenes **4a–e**



Allene	R	R[1]	Yield[a,b] (%) of azulene **7**	Yield[a,b] (%) of indene **8**	Time[c] (h)	Yield[a,d] (%) of indene **8**
4a	OMe	Me	59	1	3	50
4b	OMe	*i*-Pr	5	2	4	58
4c	OMe	Bn	9	16	5	50
4d	OMe	Ph	trace	30	3	81
4e	H	Me	26	2	7.5	50

[a] Yield of the isolated product after chromatography.
[b] Yield 30 min after the start of the experiment.
[c] Time for **4a–e** and **7a–e** to disappear (TLC monitoring).
[d] Yield after completion of thermolysis.

The higher yields of the cyclopropane-containing heterocycles **7a** and **7e** with a methyl group at the C-4b position indicate that these compounds are more stable than the 2-azabenzo[*a*]cyclopropa[*cd*]azulene **7d** substituted with a phenyl group on the same atom, which exhibits greater reactivity in the observed thermal rearrangements (Table 1).

The thermolysis products where formed as single diastereoisomers, i.e., **7** and **8**.[16] We confirmed the structures of compounds **7** and **8** by ^{1}H NMR spectroscopic analysis, which that showed the presence of single sets of signals. The ^{1}H NMR spectra of **7a–c** and **7e** were characterized by a singlet for the enamine proton H-3 at δ = 6.76–7.08 ppm, a doublet for the cyclopropane proton H-8c at δ = 1.98–2.43 ppm (^3J = 6.6 Hz), two doublets of doublets for H-1 at δ = 3.55–3.67 ($^{2,3}J$ = 14.1, 5.6 Hz) and 4.22–4.25 ($^{2,3}J$ = 14.1, 1.5 Hz), and a multiplet for H-8b at δ = 3.70–3.81. The spectra of the (epiminomethano)cyclopenta[*a*]indenes **8c** and **8d** were characterized by a singlet for H-3 at δ = 3.53–4.24 ppm, whereas those for **8a**, **8b**, and **8e** were characterized by a doublet in the same range with ^4J = 1.4–1.7 Hz. The H-9 methylene bonded to the N atom produced a doublet of doublets at δ = 2.21–2.54 ($^{2,3}J$ = 10.6, 2.2 Hz) or a doublet of doublets at δ = 2.39–2.66 ppm ($^{2,3}J$ = 10.6, 2.2 Hz) for **7a–c**

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and **7e**. The H-8a atom appeared as a doublet at δ = 3.08–3.44 ppm (^{3}J = 4.1-5.8 Hz) for **8a**, **8c**, and **8e** or as a doublet of doublets in the same range with $^{3,4}J$ = 5.6, 1.5 Hz for **8b** and **8d**.

The structures of **7a** (Figure 1) and **8a** (Figure 2) were unambiguously confirmed by X-ray crystallographic analysis.[17]

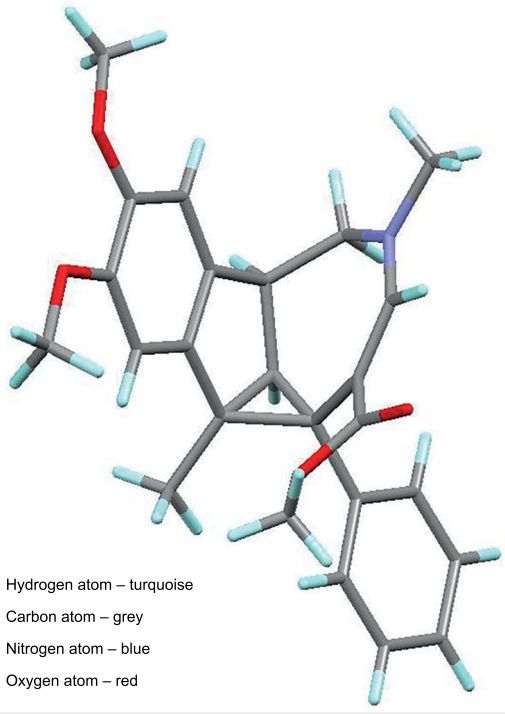


Figure 1 Single crystal X-ray analysis of compound **7a**

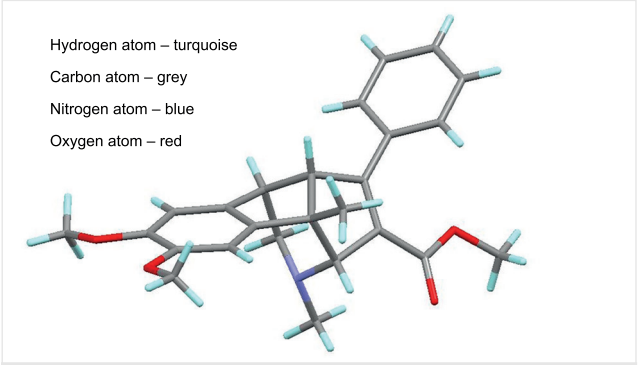
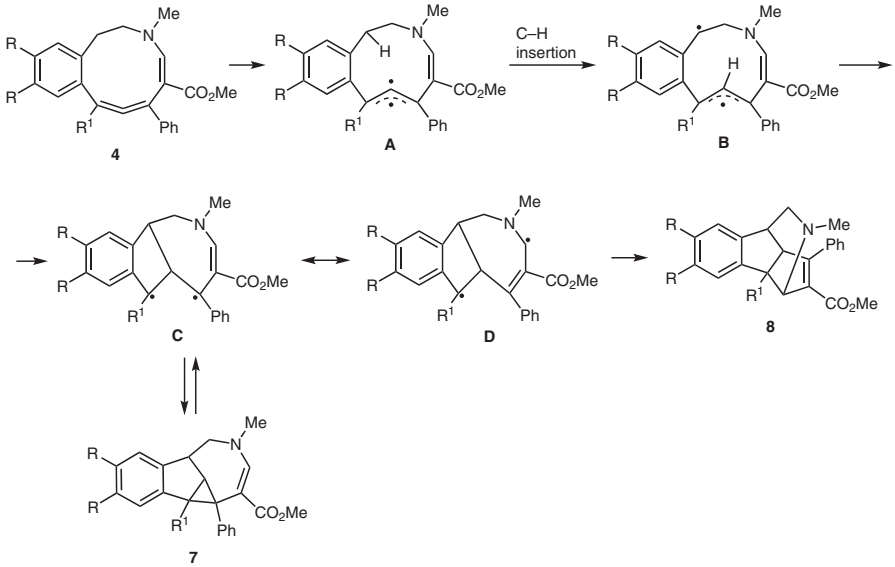


Figure 2 Single crystal X-ray analysis of compound **8a**

It is well known that allenes can be excited to form a biradical **A** that undergoes skeletal rearrangement.[18–20] Because the rearrangement of compounds **4** occurred upon heating to 150 °C, a radical mechanism can be presumed. We therefore assume that the reaction mechanism starts with formation of biradical intermediate **A** (Scheme 3).[21] In the next stage, hydrogen migration occurs from the benzyl CH_2-group to the C-7 carbon, and the cyclopropane ring **7** is formed via intermediates **B** and **C**. Further heating of the cyclopropane derivative results in its transformation into product **8** via biradical **D**.

We experimentally established that heating the isolated compound **7a** (R = OMe; R^1 = Me) gave product **8a** in 50% yield (SI and Procedure C).[22] This rearrangement leading to the thermodynamically more stable product **8a** with a cyclopentene moiety, can be considered as a retro-reaction of the cyclopropane-containing compound **7a** through C–C bond cleavage of a three-membered ring. A similar transfor-



Scheme 3 Proposed mechanism via biradical intermediates

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mation of vinylcyclopropanes into cyclopentenes has been well described in the literature.[23–25] The thermal transformation of vinylcyclopropanes into cyclopentene derivatives has been extensively reviewed by Goldschmidt and Crammer.[26]

In summary, the thermal transformations of benzazecines containing enamine and allene moieties leading to azabenzo[*a*]cyclopropa[*cd*]azulenes **7** and (epiminomethano)cyclopenta[*a*]indenes **8** was examined. The azulenes **7** have been shown to be intermediates in the synthesis of the indenes **8**.

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Supporting Information

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Primary Data

for this article are available online at https://doi.org/10.1055/s-0039-1691586 and can be cited using the following DOI: 10.4125/pd0114th.

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(13) **Methyl 8-Isopropyl-10,11-dimethoxy-3-methyl-6-phenyl-3-azabenzo[*d*]cyclodeca-4,6,7-triene-5-carboxylate** (**4b**)
Colorless crystals; yield: 438 mg (92%); mp 175–176 °C; R_f = 0.44 (EtOAc–hexane, 1:2). IR (KBr): 1956, 1055 (C=C=C), 1684 (C=O) cm^{-1}. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.61 (s, 1 H), 7.33–7.30 (m, 2 H), 7.25–7.23 (m, 2 H), 7.14–7.12 (m, 1 H), 6.88 (s, 1 H), 6.59 (s, 1 H), 4.18–4.14 (m, 1 H), 3.91 (s, 3 H), 3.85 (s, 3 H), 3.61 (s, 3 H), 3.50–3.46 (m, 1 H), 3.16 (s, 3 H), 2.94 (sept, *J* = 6.7, 1 H), 2.81–2.70 (m, 2 H), 1.35 (d, *J* = 6.7 Hz, 3 H), 1.02 (d, *J* = 6.7 Hz, 3 H). ^{13}C NMR (150 MHz, $CDCl_3$): δ = 206.0, 170.7, 147.9, 147.6, 147.4, 138.3, 129.3, 128.7, 128.2 (2 C), 126.3, 125.7 (2 C), 112.9, 111.5, 110.5, 103.5, 96.0, 56.1, 55.9, 51.8, 51.1, 45.1, 31.7, 31.0, 22.2, 21.2. LC-MS (ESI): *m/z* = 434 [M + H]$^+$. Anal. Calcd for $C_{27}H_{31}NO_4$: C, 74.80; H, 7.21; N, 3.23. Found: C, 74.95; H, 7.41; N, 3.37.
(14) **Methyl 10,11-Dimethoxy-3-methyl-6,8-diphenyl-3-azabenzo[*d*]cyclodeca-4,6,7-triene-5-carboxylate** (**4d**)
Colorless crystals; yield: 473 mg (92%); mp 210–212 °C. R_f = 0.47 (EtOAc–hexane, 1:1). IR (KBr): 1947, 1059 (C=C=C), 1690 (C=O) cm^{-1}. ^{1}H NMR (600 MHz, DMSO-d_6): δ = 7.58 (s, 1 H), 7.42–7.38 (m, 4 H), 7.34–7.29 (m, 5 H), 7.22–7.20 (m, 1 H), 6.93 (s, 1 H), 6.70 (s, 1 H), 3.89–3.85 (m, 1 H), 3.76 (s, 3 H), 3.61 (s, 3 H), 3.47 (s, 3 H), 3.29–3.25 (m, 1 H), 3.17 (s, 3 H), 3.07–3.02 (m, 1 H), 2.98–2.92 (m, 1 H). ^{13}C NMR (150 MHz, DMSO-d_6): δ = 209.1, 168.4, 148.1, 147.7, 147.2, 137.4, 136.5, 129.5, 128.8 (2 C), 128.7 (2 C), 127.4, 127.3 (2 C), 127.1, 125.9, 125.5 (2 C), 113.7, 112.8, 107.0, 104.7, 93.1, 55.6, 55.5, 50.9, 50.7, 44.6, 30.2. LC-MS (ESI): *m/z* = 468 [M + H]$^+$. Anal. Calcd for $C_{30}H_{29}NO_4$: C, 77.06; H, 6.25; N, 3.00. Found: C, 77.19; H, 6.33; N, 3.13.
(15) **General Procedure A**
A solution of the appropriate allene derivative (0.68 mmol) in dry toluene (6.0 mL) was placed in a microwave reactor and heated at 150 °C for 30 min while the progress of the reaction was monitored by TLC [EtOAc–hexane, 1:1 (**4a** and **4c**); EtOAc–hexane, 1:2 (**4b**, **4d**, and **4e**)]. The solvent was evaporated in vacuo. Products **7a–c**, **7e**, and **8a–e** were isolated by chromatography on silica gel; compounds **7a** and **7c**, **7b** and **8b**, **7e**, **8a**, **8c**, and **8e** were eluted with EtOAc–hexane (1:3, 1:5, 1:12, 1:2, 1:1, and 1:10, respectively).
(16) **General Procedure B**
A solution of the appropriate allene derivative (0.16 mmol) in dry toluene (6.0 mL) was irradiated in a microwave reactor at 150 °C for 3–7.5 h with stirring while the progress of the reaction was monitored by TLC [EtOAc–hexane, 1:1 (**4a** and **4c**); EtOAc–hexane, 1:2, (**4b**, **4d**, and **4e**)]. The solvent was evaporated in vacuo and products **8a–e** were crystallized from Et_2O.
(17) **Methyl 6,7-Dimethoxy-2,4b-dimethyl-4a-phenyl-1,2,4a,4b,8b,8c-hexahydro-2-azabenzo[*a*]cyclopropa[*cd*]azulene-4-carboxylate** (**7a**)
Colorless crystals; yield: 162 mg (59%) (Procedure A); mp 153–155 °C. R_f = 0.47 (EtOAc–hexane, 1:1). IR (KBr): 1682 (C=O) cm^{-1}. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.57–7.56 (m, 2 H), 7.26–7.24 (m, 2 H), 7.15–7.13 (m, 1 H), 6.87 (s, 1 H), 6.74 (s, 1 H), 6.50 (s, 1 H), 4.22 (dd, *J* = 14.1, 1.5 Hz, 1 H), 3.88 (s, 3 H), 3.85 (s, 3 H), 3.77–3.75 (m, 1 H), 3.58 (dd, *J* = 14.1, 5.6 Hz, 1 H), 3.44 (s, 3 H), 3.00 (s, 3 H), 2.12 (d, *J* = 6.6 Hz, 1 H), 1.71 (s, 3 H). ^{13}C NMR (150 MHz, $CDCl_3$): δ = 168.7, 147.5, 147.4, 146.7, 144.1, 140.1, 134.6, 131.1 (2 C), 127.3 (2 C), 125.3, 107.5, 106.8, 98.0, 55.8, 55.6, 51.4, 49.8, 46.5, 42.1, 39.1, 37.8, 37.0, 16.2. LC-MS (ESI): *m/z* = 406 [M + H]$^+$. Anal. Calcd for $C_{25}H_{27}NO_4$: C, 74.05; H, 6.71; N 3.45. Found: C, 74.21; H, 6.86; N, 3.55.
Methyl 5,6-Dimethoxy-2,11-dimethyl-14-phenyl-11-azatetracyclo[7.5.0.0^{2,12}.0^{3,8}]tetradeca-3,5,7,13-tetraene-13-carboxylate (**8a**)
Colorless crystals; yield: 32 mg (50%) (Procedure B); 3 mg (1%) (Procedure A); mp 149–151 °C. R_f = 0.52 (EtOAc–hexane, 1:1). IR (KBr): 1694 (C=O) cm^{-1}. ^{1}H NMR (600 MHz, $CDCl_3$): δ = 7.49–7.47 (m, 2 H), 7.41–7.35 (m, 3 H), 6.74 (s, 1 H), 6.73 (s, 1 H), 3.92

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(s, 3 H), 3.89 (s, 3 H), 3.70 (s, 3 H), 3.68 (d, *J* = 1.7 Hz, 1 H), 3.23 (d, *J* = 4.1 Hz, 1 H), 3.12–3.11 (m, 1 H), 2.66 (dd, *J* = 10.7, 2.5 Hz, 1 H), 2.48 (dd, *J* = 10.7, 2.5 Hz, 1 H), 2.22 (s, 3 H), 1.48 (s, 3 H). ^{13}C NMR (150 MHz, $CDCl_3$): δ = 166.3, 151.1, 148.3, 147.8, 137.3, 137.1, 135.1, 128.5, 128.0 (2 C), 127.9 (2 C), 124.9, 107.1, 106.9, 72.8, 67.9, 59.6, 55.8, 55.7, 51.2, 51.1, 42.6, 42.3, 18.3. LC-MS (ESI): *m/z* = 406 $[M + H]^+$. Anal. Calcd for $C_{25}H_{27}NO_4$: C, 74.05; H, 6.71; N, 3.45. Found: C, 74.28; H, 6.91; N, 3.56.

(18) CCDC 1830405 and CCDC 1579823 contain the supplementary crystallographic data for compounds **7a** and **8a**, respectively. The data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/getstructures.

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(23) **General Procedure C**
A solution of compound **7a** (0.24 mmol) in absolute toluene (6.0 mL) was irradiated in a microwave reactor at 150 °C for 2 h with stirring while the progress of the reaction was monitored by TLC (EtOAc–hexane, 1:1). The solvent was then evaporated in vacuo and product **8a** was crystallized from Et_2O; yield: 49 mg (50%).

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